

Semantic Counterfactuals Expose Branch-Identity Shortcuts in Bimanual Wire-Harness Manipulation

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DRAFT—NOT FOR SUBMISSION

Abstract—Bimanual manipulation of a branched wire harness must preserve endpoint identity as well as geometry. We introduce an exact semantic counterfactual in which cable state, dynamics, fixtures, and random seed are fixed while the observable endpoint–arm–target binding is renamed. Across 600 physical-seed–assignment episodes per policy, the full relational model succeeds in 480 (80.0%), typed ACT variants in 478 and 479, and the teacher in 478. Geometry succeeds in 123 (20.5%); every success occurs under the canonical assignment and none after the A/B swap. Geometry is lower in all six paired cells after Holm correction. Full and ablated semantic models remain tied near the teacher ceiling, showing that the benefit is not unique to message passing: it follows from preserving observable identity through representation and typed action grounding. A direct current-pose MuJoCo backend yields 40/40 successes for the relational model and 13/40 for Geometry; all Geometry successes again occur only under the canonical assignment. The evidence establishes a semantic-shortcut failure under a fixed two-branch abstraction. It does not establish 3–5 branch generalization, image-based identity perception, force control, or production performance.

Index Terms—Bimanual manipulation, deformable object manipulation, graph representations, semantic grounding, wire-harness assembly.

I. Introduction

Wire-harness automation must preserve identity while controlling shape. Two branches can be nearly identical in length and geometry yet terminate in different connectors or vehicle subsystems. Swapping them may leave a tidy, crossing-free harness that is electrically and procedurally wrong. A useful controller must therefore answer a relational question—which physical endpoint belongs to which arm and destination—before executing geometric motion.

Prior systems provide strong real-robot evidence for dense-knot recovery [3], bilateral long-cable untangling [5], learned tracing and manipulation [6], and dual-arm DLO planning in constrained 3-D environments [9], [8]. BranchBot does not claim to be the first bimanual cable system. We instead isolate a narrower failure mode that these systems do not directly measure: sensitivity to a pure renaming of branch identity when all physical evidence is held fixed.

Our contributions are:

- 1) a paired A/B intervention that exactly reuses each physical seed while changing only observable endpoint–arm–target relations;
- 2) a matched family of graph, Transformer, and typed-pointer controllers trained on balanced, episode-disjoint demonstrations, revealing which semantic

channels are sufficient rather than crediting one architecture;

- 3) exact paired statistics, semantic-invariance diagnostics, a mutually exclusive failure taxonomy, and direct closed-loop MuJoCo replication;
- 4) an explicit evidence boundary separating two-branch semantic assignment from multi-branch, vision, force, and hardware claims.

II. Related Work

a) Cable untangling and tracing.: Learned grasp features and recovery policies untangle dense non-planar knots on real cables [3]; SGTm combines RGB-D perception with bilateral motion primitives for long cables [5]. HANDLOOM learns traces and crossing classes from simulated and real RGB-D data and supports bimanual manipulation [6]. Those works provide stronger perception and hardware evidence than ours. Their public data lack matched branched endpoint–target renamings, so we do not report them as real-world semantic completion trials.

b) Dual-arm DLO control.: Coarse-to-fine and whole-body frameworks combine global planning with closed-loop control under obstacles and model uncertainty [9], [8]. Tactile or force-aware industrial harnessing further demonstrates why task-space mocap cannot stand in for contact regulation [10]. Our fixed two-branch simulator is designed for many exact counterfactual pairs rather than physical fidelity.

c) Learning representations and temporal actions.: Graph networks have modeled cable deformation [7]; graph structure alone is not novel. ACT predicts temporally chunked bimanual actions [11], while role assignment can simplify bimanual coordination [1]. We adapt ACT to the same typed state and two-pointer action interface so a continuous-output mismatch cannot create an artificial advantage for the graph policy.

III. Problem Formulation

Let $G_t = (V_t, E_t^p, E_t^s)$ contain observable cable particles, target and arm nodes. E_t^p encodes cable connectivity; E_t^s connects each semantic endpoint to its arm, two clearance waypoints, and destination. A rollout is successful only if both semantic endpoint errors fall within the difficulty-dependent tolerance, projected cable crossings are zero, and the condition is stable for four steps.

For a frozen physical seed s , intervention $z \in \{0, 1\}$ either retains or swaps the A/B names. Cable coordinates,

target coordinates, initial arms, noise, material parameters, and integration are identical. When z changes, the endpoint–arm–target relation and corresponding action slots change together. A correct equivariant policy should preserve success, while a fixed-name or position shortcut can silently manipulate the wrong endpoint.

IV. Method

The PBD harness [2] contains a ten-particle trunk and two nine-particle branches joined at one junction. Each node stores normalized position, generic type, endpoint/junction role, A/B identity, process order, state, progress, and difficulty. TopoHarness performs three message-passing layers over $\text{clip}(E^p + E^s, 0, 1)$ and produces two node-pointer heads in addition to six normalized action logits. Pointer slot A is masked to observable A targets and slot B to B targets. Selected coordinates replace the continuous position outputs; both arms retain their assigned endpoints.

The Geometry model has the same hidden width and training schedule but removes adjacency, endpoint/junction identity, process order, and A/B features. Three graph ablations retain only physical edges (leaving unary identity), retain only semantic edges, or retain both relations while removing unary ID/order features. The ACT-style baseline is an independent compact implementation with a CVAE Transformer, eight-action chunks, temporal ensembling, and the same data. ACT-Pointer adds legal A/B candidate sets at action grounding. ACT-RelPool additionally averages fixed one-hop semantic-neighbor features before its Transformer; it does not use learned graph propagation. Thus the suite tests several distinct ways of carrying semantic identity to the action, not merely graph versus non-graph architecture.

V. Experimental Protocol

The raw corpus contains 420 complete teacher rollouts and 43,258 sampled states. Because the teacher can fail on difficult cable states, model fitting uses a frozen successful-only subset of 160/40/80 training/validation/test episodes (25,740 states; SHA-256 prefix 2e52208d2f41). Every split is balanced across difficulty and A/B assignment; whole episodes and physical seeds are disjoint. The final evaluation starts at seed 66,000,000, outside all corpus seed ranges, and contains 100 new physical seeds at each difficulty 0.2, 0.5, 0.8. Both semantic assignments are executed for every seed.

All learned graph policies use 64 hidden units and three layers. ACT variants use 48 hidden units, one encoder/decoder layer, and eight-step chunks. Models see identical demonstrations within their family; model selection uses only validation loss. We report Wilson intervals, per-difficulty and per-assignment success, endpoint errors, crossings, motion length, failures, and paired exact McNemar tests with Holm family-wise correction. A single training seed per model remains a limitation; paired evaluation seeds quantify rollout, not training, uncertainty.

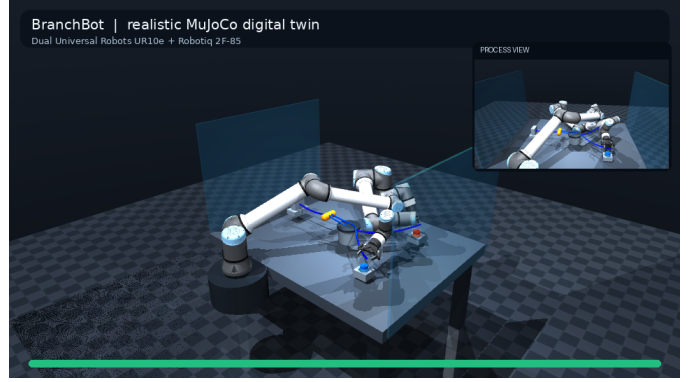


Fig. 1. BranchBot visual digital twin: two UR10e arms and Robotiq 2F-85 grippers around a native MuJoCo trunk–junction–branch cable. Robot meshes are collision-disabled visual IK followers and do not alter cable physics.

TABLE I

Frozen PBD counterfactual. Each policy receives 300 physical seeds under both semantic assignments (600 episodes).

Policy	Succ.	Rate [95% CI]
TopoHarness	480/600	80.0 [76.6,83.0]
No semantic edges	477/600	79.5 [76.1,82.5]
No physical edges	478/600	79.7 [76.3,82.7]
Relations, no ID/order	481/600	80.2 [76.8,83.2]
Geometry	123/600	20.5 [17.5,23.9]
ACT direct	432/600	72.0 [68.3,75.4]
ACT-Pointer	478/600	79.7 [76.3,82.7]
ACT-RelPool	479/600	79.8 [76.4,82.8]
Scripted teacher	478/600	79.7 [76.3,82.7]
Random	0/600	0.0 [0.0,0.6]

VI. Results

a) Paired semantic intervention.: The frozen main table contains 480 relational successes and 123 Geometry successes. TopoHarness differs on only 2 of 300 canonical/swapped physical pairs (99.3% outcome agreement). Geometry obtains all 123 successes in the canonical condition and 0/300 after the A/B swap; its paired outcome agreement is 59.0%. Geometry is lower than TopoHarness in every difficulty–assignment cell after Holm correction, with the largest adjusted value 2.56×10^{-3} . Untyped ACT reaches 432/600 and exhibits 15 canonical-success/swapped-failure pairs versus one in the reverse direction. At difficulty 0.8 it is lower than TopoHarness for both assignments (adjusted $p = 5.94 \times 10^{-3}$ and 4.81×10^{-6}). ACT-Pointer reaches 478 and ACT-RelPool 479, both near the scripted teacher’s 478. The exact paired tests avoid treating the two semantic assignments of one physical seed as independent.

b) Ablations and interpretation.: Removing semantic edges (while retaining unary identity), removing physical edges (while retaining semantic identity), or removing unary ID/order (while retaining semantic relations) yields 477, 478, and 481 successes, respectively, versus 480 for TopoHarness. Typed ACT also reaches the same band. The data therefore reject a unique message-passing

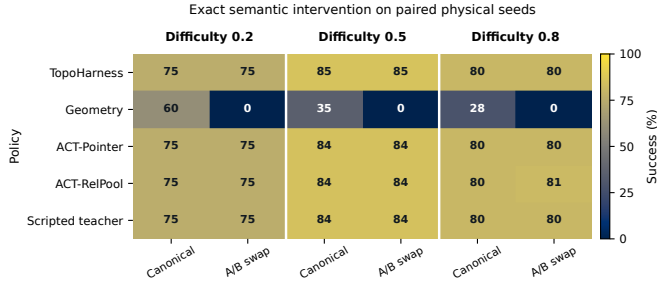


Fig. 2. Success under the exact A/B intervention (100 paired physical seeds per cell). Both columns share identical physics; only semantic binding is renamed.

TABLE II

Direct current-pose MuJoCo observation/task-space-control audit. The detailed arms are visual IK twins, not force controllers.

Policy	Succ.	Rate [95% CI]
TopoHarness	40/40	100.0 [91.2,100.0]
Geometry	13/40	32.5 [20.1,48.0]
Scripted teacher	40/40	100.0 [91.2,100.0]

explanation: multiple semantic channels are sufficient when the two simultaneous commands remain type-bound. The unsupported alternative is geometry alone, which learns a canonical left/right shortcut and collapses under renaming.

c) Cross-physics closed loop.: For each of 20 paired fixture and initial-gripper perturbations, the graph is rebuilt from current MuJoCo cable bodies, sites, and mocap poses at every control step. Both policy outputs directly move the two task-space grippers; no PBD trajectory, stage event, or success label is replayed. TopoHarness completes 40/40 semantic trials and Geometry completes 13/40. TopoHarness is 20/20 under each assignment; all 13 Geometry successes are canonical and its swapped score is 0/20. This replicates the semantic-shortcut diagnosis under a second deformable backend [4], but does not validate joint torque or contact-force control.

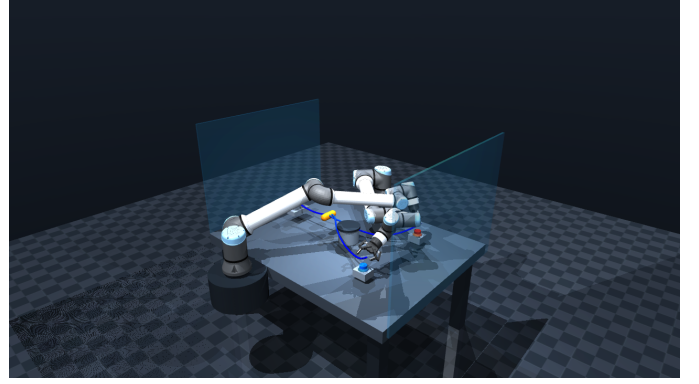


Fig. 3. Representative direct-observation MuJoCo final state. Native cable endpoint–fixture distances define success; the robot mesh only visualizes task-space commands.

VII. Limitations and Claim Boundary

The central intervention contains exactly two branches and two labels. It does not establish permutation-equivariant control for three to five branches, sequential regrasp planning, or non-symmetric branch materials. PBD is planar; its crossing count is a projection metric. MuJoCo uses 3-D elastic cables but task-space mocap grippers and collision-disabled visual robots. No experiment uses raw camera images, detects branch identity under occlusion, measures contact force, or runs on hardware. HANDLOOM is relevant public perception data but lacks the matched branched semantic labels required by our causal test, so it is not counted as real-world completion evidence. These gaps keep the manuscript a research draft.

VIII. Ethical and Societal Considerations

The study uses synthetic robot trajectories and no personal data. Harness automation could reduce repetitive ergonomic strain but may also displace manual assembly work. The simulated controller is not safety certified and must not command a production cell without guarding, risk assessment, force limits, and qualified human supervision. We therefore expose unsuccessful episodes and avoid translating simulator rates into labor or financial claims.

IX. Reproducibility

The package includes both source and successful-only datasets, whole-episode split manifests, hashes, frozen checkpoints, exact seeds, per-episode CSV/JSON, statistical audit scripts, plotting code, MJCF, visual-asset licenses, and a one-command build. The ACT baseline is an independent adaptation and is not presented as upstream ACT code. An audit recomputes counts, Wilson inputs, semantic pairs, exact tests, checkpoint hashes, and PBD/MuJoCo seed separation before the PDF is copied to the release directory.

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